

The Faddeev-Popov Determinant

Perimeter Scholars International, Quantum Field Theory II, Tutorial 9; 10 worked parts.

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Quantum Field Theory

Question prompts paraphrased from the official Perimeter PSI Online sheet; the equations and notation follow the source.

1. The Faddeev–Popov Determinant

As a finite-dimensional model of an unfixed gauge zero mode, take

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A = \begin{pmatrix} a \\ b \end{pmatrix}. \quad (1.1)$$

Then

$$I = \int_{-\infty}^{\infty} dA e^{-A^T D A} = \int_{-\infty}^{\infty} da db e^{-a^2} \quad (1.2)$$

is ill-defined because one factor is the volume of the real line.

For the general gauge-invariant integral

$$I = \int \mathcal{D}A e^{iS(A)}, \quad (1.3)$$

assume that $A \mapsto A_g$ belongs to a gauge group $\mathcal{G}$ and satisfies

$$\mathcal{D}A = \mathcal{D}A_g, \quad S(A) = S(A_g). \quad (1.4)$$

Define the Faddeev–Popov determinant by

$$1 = \Delta(A) \int \mathcal{D}g \delta[f(A_g)], \quad \int \mathcal{D}g = \text{Vol}(\mathcal{G}). \quad (1.5)$$

a) Prove that $\Delta(A)$ is gauge invariant.

Solution. Under a gauge transformation

$$A \mapsto A_h, \quad (A_h)_g = A_{gh}, \quad \Delta(A) \mapsto \Delta(A_h), \quad (1.6)$$

we have

$$1 = \Delta(A_h) \int \mathcal{D}g \delta[f(A_{gh})]. \quad (1.7)$$

Formally, the functional integral over the gauge group is defined as

$$\mathcal{D}g = \prod_x d\mu_H(g(x)), \quad (1.8)$$

where $d\mu_H$ is Haar measure. It is the group theoretical analogue to the normal translational invariant measure, $dx = d(x+a)$. In group theory, the abelian operation $+$ is replaced by actions like right or left actions. In our present case, we take the Haar measure to be right invariant, namely

$$\mathcal{D}g = \mathcal{D}(gh). \quad (1.9)$$

Therefore, we have

$$1 = \Delta(A_h) \int \mathcal{D}(gh) \delta[f(A_{gh})] = \Delta(A_h) \int \mathcal{D}g' \delta[f(A_{g'})] = \Delta(A) \int \mathcal{D}g \delta[f(A_g)] \implies \Delta(A) = \Delta(A_h). \quad (1.10)$$

b) Insert the identity above and show that

$$I = \text{Vol}(\mathcal{G}) \int \mathcal{D}A e^{iS(A)} \Delta(A) \delta[f(A)]. \quad (1.11)$$

Solution. Inserting the identity into the path integral, we find

$$I = \int \mathcal{D}A e^{iS[A]} \Delta(A) \int \mathcal{D}g \delta[f(A_g)]. \quad (1.12)$$

Using gauge invariant of the action and the integral measure, $\mathcal{D}A = \mathcal{D}A_g$, we find

$$I = \left(\int \mathcal{D}g \right) \int \mathcal{D}A_g e^{iS(A_g)} \Delta(A_g) \delta[f(A_g)] = \text{Vol}(\mathcal{G}) \int \mathcal{D}A e^{iS(A)} \Delta(A) \delta[f(A)]. \quad (1.13)$$

Next use the zero-dimensional toy theory

$$Z = \int_{-\infty}^{\infty} dx dy e^{-S(x,y)}, \quad S(x,y) = (x-y)^2. \quad (1.14)$$

c) Identify the global “gauge” symmetry of this zero-dimensional action.

Solution. The toy theory is clearly invariant under translation, $x \rightarrow x + a, y \rightarrow y + a$.

d) Describe the full configuration space $\mathcal{A}$ and its gauge orbits $\mathcal{G}$.

Solution.

$$\mathcal{A} = \{x, y \in \mathbb{R}\} = \mathbb{R}^2, \quad \mathcal{G} = (\mathbb{R}, +) \implies \mathcal{A}/\mathcal{G} \cong \mathbb{R}. \quad (1.15)$$

The divergent partition function can be factorized as

$$Z = \left(\int da \right) X, \quad (1.16)$$

where a parametrizes the gauge group and X is independent of a .

e) Find new variables that make this factorization explicit. As an additional challenge, suggest a modified model whose gauge-group volume is finite.

Solution. Perform the change of variable,

$$a = \frac{x-y}{\sqrt{2}}, \quad b = \frac{x+y}{\sqrt{2}}, \quad (1.17)$$

which gives

$$Z = \left(\int da \right) \int db e^{-b^2}. \quad (1.18)$$

f) Choose a gauge slice for computing X and give a condition that characterizes it; the choice need not be unique. Denote the condition by $f(x, y) = 0$.

Solution. It is easiest if we choose the gauge slice $f(x, y) = y = 0$.

g) Show that

$$Z = \int_{-\infty}^{\infty} da \int_{-\infty}^{\infty} dx dy J \delta(f(x, y)) e^{-(x-y)^2}, \quad (1.19)$$

where the Jacobian J is independent of a .

Solution. We have

$$1 = \int df(x, y) \delta(f(x, y)) = \int df(x + a, y + a) \delta(f(x + a, y + a)) = \int da J \delta(f(x + a, y + a)). \quad (1.20)$$

For our particular gauge choice, this is

$$1 = \int da \delta(y + a). \quad (1.21)$$

Inserting into the partition function, we find

$$Z = \int_{-\infty}^{\infty} da \int_{-\infty}^{\infty} dx dy J \delta(f(x, y)) e^{-(x-y)^2} \quad (1.22)$$

by "gauge" invariance of the action and the measure.

Replace the gauge slice $f = 0$ by $f = c$, with constant $c \neq 0$, so that

$$Z = \int da \int dx dy J \delta(f(x, y) - c) e^{-(x-y)^2}. \quad (1.23)$$

h) Although the last expression displays c explicitly, prove that Z is independent of c .

Solution. Since

$$1 = \int df(x, y) \delta(f(x, y)) = 1 = \int df(x, y) \delta(f(x, y) - c), \quad (1.24)$$

the rest follows straightforwardly from above.

i) Average over c with the Gaussian weight

$$\int_{-\infty}^{\infty} dc e^{-c^2} \quad (1.25)$$

and derive

$$Z = \frac{1}{\sqrt{\pi}} \int da \int dx dy J e^{-f^2} e^{-(x-y)^2}. \quad (1.26)$$

Explain how the factor e^{-f^2} supplies a gauge-fixing term.

Solution.

$$Z = \frac{1}{\sqrt{\pi}} \int dc e^{-c^2} \int da \int dx dy J \delta(f(x, y) - c) e^{-(x-y)^2} = \frac{1}{\sqrt{\pi}} \int da \int dx dy J e^{-f^2} e^{-(x-y)^2}. \quad (1.27)$$

This factor is largest when $f = 0$ and exponentially suppresses configurations for which $f \neq 0$. It therefore imposes the same gauge condition softly.

j) Exponentiate the remaining Jacobian by representing it as an integral over Grassmann "ghost" variables.

Solution.

$$J = \int d\bar{c} dc e^{-\bar{c} J c}. \quad (1.28)$$